

A LINEAR ERROR BOUND FOR ERDŐS PROBLEM 272

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ABSTRACT. Let $t(N)$ be the largest size of a family of subsets of $\{1, \dots, N\}$ whose intersections of distinct members are nonempty arithmetic progressions. We prove

$$t(N) = \frac{N^2}{2} + O(N).$$

The proof gives $t(N) \leq N^2/2 + 30000N$ for all sufficiently large N . After removing at most $2048N$ members, a family of size at least $N^2/2$ has a point c such that either every member contains c , or every member avoiding c is an interval containing a long common core. A matching of progressions to primitive pairs, together with private witnesses outside a reserved set of exterior pairs, then gives the linear error. The endpoint stability argument uses a finite normalization of coprime-pair counts, preserving the quadratic deficit through summation over all steps. We supply a complete Lean 4 formalization of the Szabó strong variant of Erdős problem 272, with its exact target type and an audit of its axioms.

1. INTRODUCTION

Write $[N] = \{1, \dots, N\}$. A family $\mathcal{F} \subseteq 2^{[N]}$ is *well-intersecting* if $A \cap B$ is a nonempty arithmetic progression whenever $A, B \in \mathcal{F}$ and $A \neq B$. Singletons and two-element sets are arithmetic progressions. Let $t(N)$ be the maximum cardinality of such a family. The restriction concerns intersections of distinct members; a member of the family need not itself be an arithmetic progression.

Simonovits and Sós studied this extremal problem in [1]. Szabó proved the estimate

$$t(N) = N^2/2 + O(N^{5/3}(\log N)^3)$$

and asked whether the error can be reduced to $O(N)$ [2]. This is the Szabó strong variant of Erdős problem 272 in the Formal Conjectures library [4].

Theorem 1.1. *There is an integer N_0 such that, for every $N \geq N_0$,*

$$\binom{N}{2} + 1 \leq t(N) \leq \frac{N^2}{2} + 30000N. \quad (1.1)$$

Consequently $t(N) - N^2/2 = O(N)$ as $N \rightarrow \infty$.

The coefficient 30000 is chosen for convenient bookkeeping. The argument does not determine the optimal coefficient of N , an exact formula for $t(N)$, or the structure of every extremal family. In particular, the structural reduction below allows a family without a common point.

Yang’s recent preprint [3] studies exact values and proves an exact upper bound for families with a common element, using private pairs and a staircase counting argument. These rooted ideas overlap the starting point of the present proof. We use weaker rooted bounds with absolute linear errors, and prove the additional reductions needed for arbitrary families. The complete formal development includes the combinatorial estimates used in those reductions; they are not introduced as additional assumptions.

The paper is a mathematical account of a proof accompanied by a complete formal certificate. We explain the finite counting constructions and the assembly of the argument, and identify the principal

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declarations in the Lean file. The certificate contains the detailed finite-set case analyses, floor-function estimates, and uniform threshold arguments. Section 10 specifies its exact statement and verification scope.

1.1. The lower bound. Fix $c \in [N]$, and take every subset containing c of size at most three. Two distinct such subsets intersect in one or two points, so this family is well-intersecting. Its cardinality is

$$1 + (N - 1) + \binom{N - 1}{2} = \binom{N}{2} + 1.$$

Thus only the upper bound in Theorem 1.1 requires proof.

Lean declaration: `lower_bound`.

1.2. Outline of the upper bound. We shall use the following structural statement. Reflection means applying the map $x \mapsto N + 1 - x$ to every set; it preserves cardinalities and the well-intersecting property.

Theorem 1.2 (Structural reduction). *For all sufficiently large N , every well-intersecting family $\mathcal{F}$ with $|\mathcal{F}| \geq N^2/2$ admits a well-intersecting family $\mathcal{G}$ with*

$$|\mathcal{F}| \leq |\mathcal{G}| + 2048N \tag{1.2}$$

and one of the following properties:

- (1) *all members of $\mathcal{G}$ contain a common point c ;*
- (2) *there are integers $c < L \leq R \leq N$ such that*

$$256(L + N - R)(\lfloor \log_2 N \rfloor + 1) \leq N, \tag{1.3}$$

and every member of $\mathcal{G}$ avoiding c is an interval $[a, b]$ satisfying $c < a \leq L \leq R \leq b \leq N$.

Here $\mathcal{G}$ can be obtained by possibly reflecting a subfamily of $\mathcal{F}$.

The first alternative has $|\mathcal{G}| \leq \binom{N}{2} + 20001N$. In the second, we reserve one exterior pair for each interval avoiding c , and match the members containing c into the pairs left over, with at most $20001N$ exceptions including the small members. At most $2N$ boundary intervals are set aside. This gives

$$|\mathcal{G}| \leq \binom{N}{2} + 20003N. \tag{1.4}$$

Sections 3 and 9 prove these counting bounds. Sections 4–8 establish the structural reduction. The crucial quantitative input is an endpoint deficit of the form

$$|\mathcal{G}| \leq \frac{N^2}{2} - \frac{L^2 + (N - R)^2}{100} + O(N \log^3 N).$$

Its role is to force (1.3); the logarithmic error is not the final error in the extremal bound.

2. ARITHMETIC HULLS AND PRIVATE WITNESSES

For a finite integer set T with at least two elements, let $\text{ah}(T)$ be the shortest arithmetic progression containing T . If $a = \min T$ and $d = \gcd\{x - a : x \in T\}$, then

$$\text{ah}(T) = \{a, a + d, \dots, \max T\}.$$

Every arithmetic progression containing T contains $\text{ah}(T)$.

Definition 2.1. A finite set is *crooked* if it is not an arithmetic progression. For $c \in S$, a two-element subset $\{u, v\} \subseteq S \setminus \{c\}$ is an *intrinsic witness at c* if

$$\text{ah}(\{c, u, v\}) \not\subseteq S. \tag{2.1}$$

Equivalently, no arithmetic progression contained in S contains c, u, v .

Lemma 2.2 (Privacy). *If $S \in \mathcal{F}$ has an intrinsic witness $\{u, v\}$ at c , no other member of $\mathcal{F}$ contains c, u, v . In particular, witnesses at a fixed root are distinct for distinct members containing that root.*

Proof. If $T \neq S$ contained the three points, the arithmetic progression $S \cap T$ would contain their arithmetic hull. This would contradict (2.1). $\square$

Lemma 2.3 (Root closure). *Every crooked set S containing c has an intrinsic witness at c .*

Proof. Translate c to zero and suppose every triple through zero has its hull contained in S . If necessary reflect so that S has a positive element. Starting with a positive $g \in S$, replace g successively by $\gcd(g, |x|)$ for the nonzero $x \in S$. Each new positive gcd lies in the hull of $\{0, g, x\}$ and hence in S . At the end the gcd $d > 0$ of all elements of S belongs to S . The hulls through $0, d$ and the two extreme points fill every multiple of d between those extremes. All elements of S are such multiples, so S is an arithmetic progression. $\square$

We also need witnesses attached to prescribed nearby points. If $u < v$ are consecutive elements of a crooked set S , then some $z \in S$ has $\text{ah}(\{u, v, z\}) \not\subseteq S$. Indeed, progressions contained in S in which u, v remain consecutive all have step $v - u$. Their union is again a progression of that step. If each triple had an arithmetic cover in S , that union would contain every point of S .

The following uniform version permits a witness to meet a prescribed set. Put, for $N \geq 1$,

$$H_N = \sum_{d=1}^N \frac{1}{d}, \quad J_N = \lfloor \log_2 N \rfloor + 1. \quad (2.2)$$

Lemma 2.4 (Restricted witnesses). *Suppose all members of a crooked well-intersecting family $\mathcal{C}$ contain $c \in [N]$ and meet $P \subseteq [N] \setminus \{c\}$. After discarding at most $48NJ_N^3$ members, one can choose distinct intrinsic witnesses $\{p, x\}$ with $p \in P$.*

Proof. Choose $p \in S \cap P$ for each S . If there is a witness through c, p , use it. Otherwise every triple c, p, z has an arithmetic cover in S . Write $T = S \cap [\min(c, p), \max(c, p)]$ for the trace. Every cover has, between c and p , a full progression of step dividing $|p - c|$.

If T is crooked, take an intrinsic witness $\{x, y\}$ in T at c . It is also a witness in S : any cover in S can be intersected with the trace interval. Covers of c, p, x and c, p, y provide two progression traces with steps $d, e \mid |p - c|$. Their union has no arithmetic cover in S , since it contains c, x, y . The code (p, d, e) is private: another member containing both traces would give such a cover by intersection with S . These sets number at most $\sum_{p=1}^N \tau(|p - c|)^2$.

If T is a progression of step d , every triple cover has a trace whose step is a multiple of d . Hence every $z \in S$ satisfies $d \mid |z - c|$. The neighbor a of c toward p is at distance d . The consecutive-point observation preceding the lemma gives a witness $\{a, z\}$ in S . It is determined by z , the side of a relative to c , and the divisor d of $|z - c|$. Privacy bounds this second class by $2 \sum_{z=1}^N \tau(|z - c|)$.

Use the convention $\tau(0) = 0$ in these sums and put

$$E_N(c) = \sum_{q=1}^N \tau(|q - c|)^2 + 2 \sum_{q=1}^N \tau(|q - c|).$$

Each positive distance occurs at most twice, and $\tau(m) \leq \tau(m)^2$, so $E_N(c) \leq 6 \sum_{m=1}^N \tau(m)^2$. Expanding the second moment by pairs of divisors, and writing $a = gu, b = gv$ with $(u, v) = 1$, gives

$$\sum_{m=1}^N \tau(m)^2 = \sum_{\substack{a, b \geq 1 \\ [a, b] \leq N}} \left\lfloor \frac{N}{[a, b]} \right\rfloor \leq N \sum_{guv \leq N} \frac{1}{guv} \leq NH_N^3.$$

Here $[a, b]$ denotes the least common multiple. Finally $H_N \leq 2J_N$ gives $E_N(c) \leq 48NJ_N^3$. All retained witnesses are distinct by privacy. $\square$

Lean declaration: `witnessExceptionBound_le_natLog`.

The quantity $E_N(c)$ is the certificate's `witnessExceptionBound`; throughout,

$$0 \leq E_N(c) \leq 48NJ_N^3. \quad (2.3)$$

The witnesses retained in Lemma 2.4 are intrinsic, so their privacy holds in the original well-intersecting family, including its progression members.

3. MATCHING PROGRESSIONS AND COUNTING ROOTED FAMILIES

A progression containing c and at least four points has a unique form

$$P(c, d; l, r) = \{c + jd : -l \leq j \leq r\}, \quad d \geq 1, \quad l, r \geq 0, \quad l + r \geq 3. \quad (3.1)$$

Call a pair of nonzero integers $u < v$ *primitive* if $\gcd(|u|, |v|) = 1$. We will assign to almost every cell (l, r) a primitive pair in $[-l, r] \setminus \{0\}$ such that $\{0, u, v\}$ is not a three-term progression.

Lemma 3.1 (Uniform progression matching). *There is a partial injective assignment from the cells $\{(l, r) \in \mathbb{N}^2 : l + r \geq 3\}$ to primitive pairs (u, v) such that*

$$-l \leq u < v \leq r, \quad uv \neq 0, \quad u + v \neq 0, \quad 2u \neq v, \quad u \neq 2v.$$

At most 20000 cells in any finite collection are unassigned.

Proof. Write $\Phi(m) = \sum_{b=1}^m \varphi(b)$. The elementary estimate

$$\Phi(m) \geq \frac{37}{144}m^2 \quad (3.2)$$

is enough. To see it, a noncoprime pair in $[m]^2$ either has both entries even or has both divisible by some odd integer at least three. Therefore the fraction of such pairs is at most

$$\frac{1}{4} + \sum_{k \geq 0} \frac{1}{(2k+3)^2} \leq \frac{35}{72}.$$

The last inequality follows by separating $1/9$ and bounding the remaining tail by the integral of x^{-2} over consecutive intervals of length two. The number of coprime pairs is at most $2\Phi(m)$, proving (3.2).

If $\gcd(l, r) = 1$, assign (l, r) to $(-l, r)$. The condition $l + r \geq 3$ ensures that this pair is nonzero and is not symmetric about zero. For the other cells with $r \leq l$, use negative pairs $(-b, -a)$ with $1 \leq a < b \leq l$, $b \geq 3$ and $\gcd(a, b) = 1$. The available pairs through level m number $\Phi(m) - 2$. The noncoprime cells in the triangle $0 \leq r \leq l \leq m$ number at most

$$\frac{(m+1)(m+2)}{2} - \Phi(m).$$

By (3.2), the former quantity is at least the latter for $m \geq 128$. For smaller m , a reserve of 10000 dummy values suffices. The neighborhoods are finite and nested by l . For any finite set of cells, its largest l therefore verifies Hall's inequality. The infinite form of Hall's theorem for finite neighborhoods gives an injection with at most 10000 dummy values. Treat $l < r$ by reflection, using positive pairs and a separate reserve of 10000 values.

The three types of real pairs have different sign patterns and cannot collide. A primitive positive pair with $b \geq 3$ cannot have $b = 2a$: coprimality would force $(a, b) = (1, 2)$. The corresponding negative case is identical. This verifies all the asserted inequalities. $\square$

Lean declaration: `exists_progression_pair_matching`.

After scaling by d and translating by c , a real assigned pair has $\gcd(|u-c|, |v-c|) = d$. Pairs assigned for different steps are consequently distinct. Since $d \leq N$ for a progression with at least four points in $[N]$, the total number of dummy assignments is at most $20000N$.

Proposition 3.2 (Rooted bound). *If every member of a well-intersecting family $\mathcal{F}$ contains c , then*

$$|\mathcal{F}| \leq \binom{N}{2} + 20001N. \quad (3.3)$$

Proof. There are at most N members of size at most two. Give a three-element member the pair obtained by deleting c . Give a crooked member of size at least four an intrinsic witness at c . Finally apply Lemma 3.1 to the progression members of size at least four. Witness privacy excludes collisions with another member's assigned pair. A pair from a three-element progression cannot be a pair from Lemma 3.1, because the latter forms a nonprogression with c . The remaining three-element cases are covered by privacy. Thus all real assignments are distinct subsets of $[N]$ of size two. There are at most $20000N$ dummy assignments and N small members. $\square$

Lean declaration: `common_point_card_le`.

4. A COMMON POINT FOR THE SPARSE CROOKED MEMBERS

This section records the uniform estimates that lead to the common-root reduction. Their constants are kept explicit so that the eventual quantifiers do not depend on the family.

Lemma 4.1 (Dense crooked members). *For $N \geq 64$, a well-intersecting crooked family with $16|S| > N$ for every member S has size at most $2048N$.*

Proof. Partition $[N]$ into the blocks specified by $\lfloor x/32 \rfloor$. Every point of S except the least point in its occupied block is the larger endpoint of a consecutive pair of S with gap at most 32. The number of occupied blocks is at most $\lfloor N/32 \rfloor + 1$. Under $16|S| > N$ and $N \geq 64$, this gives at least $N/64$ such pairs.

For fixed consecutive endpoints $u < v$, at most N crooked members can have u, v consecutive. Indeed, fix such a member S . Each intersection $S \cap T$ with another such member is a progression of step $v - u$ containing u, v . Their union is a progression of this same step, or is empty if there is no other member. It cannot equal the crooked set S . A point left over belongs to no other member of this class. These private points are distinct, proving the bound N . There are at most $32N$ possible pairs $u < v$ with $v - u \leq 32$. Double-counting pairs and members yields $|\mathcal{F}| N/64 \leq 32N^2$. $\square$

Lean declaration: `dense_crooked_card_le`.

Lemma 4.2 (Progression families). *A well-intersecting family consisting entirely of arithmetic progressions satisfies*

$$|\mathcal{A}| \leq \frac{5}{12}N^2 + N \log N + 20N + 3 \quad (N \geq 1). \quad (4.1)$$

Proof. For a fixed step d , nonempty pairwise intersections put all progressions in the same residue class modulo d . They are intervals in that grid, so they have a common grid point. Counting their left and right endpoints gives the bound $N^2/(4d^2) + N/d + 1$ for the main part of the step class. Summing uses $\sum_{d=1}^N d^{-2} \leq 5/3$ and $H_N \leq 1 + \log N$. The members with at most three elements contribute at most $18N + 3$. To see this, choose one nonempty such member S_0 . Every other member of the small subfamily meets one of its at most three points. Through any fixed point, a progression of size two or three is specified by its size, the point's index, and a step in $[N]$, giving at most $6N$ possibilities, plus one singleton. If no nonempty member exists there is at most one set. Adding this to the large-step count proves (4.1). $\square$

Lean declaration: `progression_family_card_le`.

Let $D_N = \max_{1 \leq m \leq N^2} \tau(m)$, with $D_0 = 0$. The elementary divisor estimate gives $D_N = N^{o(1)}$. The following finite inequality is the form of the small/medium-member argument used here.

Lemma 4.3 (No common point). *Let C be a well-intersecting crooked family with empty total intersection and $|S| \leq s$ for each $S \in C$. If $K \geq 1$ and $16J_N D_N^2 + D_N \leq K$, then*

$$|C| K \leq (sN + 4(16K)^2 N + (32K + 1)48NJ_N^3)K + 1200J_N^2 N^2. \quad (4.2)$$

Finite counting argument. Split at size $16K$. The small members are counted by exposing a member of size at most s and applying restricted witnesses at its points. If the members retained through a point have a further common point, use a member avoiding that point. The endpoint and exceptional classes give

$$sN + 4(16K)^2 N + (32K + 1)48NJ_N^3.$$

This is the certificate's small-family estimate; it only requires empty total intersection for the ambient crooked family, not for each part.

For the large part use the bands $16jK + 1 \leq |S| \leq 16(j + 1)K$, $j \geq 1$. The trace construction in Lemma 2.4 has the following alternative: a member supplies many intrinsic witnesses through a selected point, or contains a long progression core with a short remainder. For the first class, privacy bounds the total number of witnesses by the available pairs. For the second, group maximal cores by step, residue, and starting window. Two cores in one class have a long common prefix. A shared remainder point would force an arithmetic intersection extending a maximal core, so the relevant remainders are disjoint. Counting windows and steps gives

$$|C_j| jK \leq 600J_N N^2. \quad (4.3)$$

The condition on D_N ensures that the divisor-coded trace exceptions are smaller than the chosen cutoff in every band. Summing (4.3) over $1 \leq j \leq N$ gives $|C_{>16K}| K \leq 600J_N N^2 H_N \leq 1200J_N^2 N^2$. Combining this with the small-family count proves (4.2). The finite trace alternative, window count, and both class bounds are proved in the companion file before the declaration below. $\square$

Lean declaration: `crooked_no_common_point_card_mul_le`.

Proposition 4.4. *For all sufficiently large N , a well-intersecting crooked family with $16|S| \leq N$ for all members and $N^2 \leq 14|C|$ has a common point in $[N]$.*

Proof. Take $K = \lfloor \sqrt{\lfloor \sqrt{N} \rfloor} \rfloor$ in Lemma 4.3, with $s = \lfloor N/16 \rfloor$. The divisor estimate implies $16J_N D_N^2 + D_N \leq K$ eventually. After division by K , the error terms are

$$O(K^2 N) + O(KNJ_N^3) + O(J_N^2 N^2 / K) = o(N^2).$$

An empty total intersection would therefore imply $|C| \leq N^2/16 + o(N^2)$, contradicting $|C| \geq N^2/14$. All eventual estimates are uniform in C . $\square$

Lean declaration: `eventually_sparse_crooked_common_point`.

If $|\mathcal{F}| \geq N^2/2$, Lemma 4.2 implies that the crooked members number at least $N^2/12 - O(N \log N)$. Removing the dense crooked members costs at most $2048N$ by Lemma 4.1. For large N , at least $N^2/14$ sparse crooked members remain. Proposition 4.4 supplies a common point c for them. Retaining all progression members gives a subfamily $\mathcal{G} \subseteq \mathcal{F}$ such that

$$|\mathcal{F}| \leq |\mathcal{G}| + 2048N, \quad c \in S \text{ for every crooked } S \in \mathcal{G}. \quad (4.4)$$

Lean declaration: `eventually_common_crooked_reduction`.

5. EXCLUDING THE STRADDLING CASE

Fix a family $\mathcal{G}$ and a root c as in (4.4). A member avoiding c is necessarily a progression. It *straddles* c if it contains a point on each side of c . Such a progression has step at least two and lies in a residue class different from that of c modulo its step.

Lemma 5.1 (Straddling gap). *For all sufficiently large N , if every member of $\mathcal{G}$ avoiding c straddles c and at least one such member exists, then*

$$|\mathcal{G}| \leq \frac{17}{36}N^2. \quad (5.1)$$

Proof. There are two cases, with uniform errors $O(NJ_N^3)$. If all avoiding progressions have step two, fix one of them. Every rooted member meets the parity class opposite to c . The matching construction of Section 3, restricted to pairs meeting that class, gives at most $3N^2/8 + O(NJ_N^3)$ rooted members. The avoiding step-two progressions contribute at most $N^2/16 + O(N)$. Thus

$$|\mathcal{G}| \leq \frac{7}{16}N^2 + O(NJ_N^3).$$

If some avoiding progression has step at least three, every rooted crooked member meets that progression. Restricted witnesses and the joint interval count put the crooked members together with the rooted intervals below $5N^2/18 + O(NJ_N^3)$. Every interval must contain c , because a straddling interval contains c . The remaining, noninterval progressions contribute at most $N^2/6 + O(N \log N)$, by the endpoint count summed over steps at least two. Hence

$$|\mathcal{G}| \leq \frac{4}{9}N^2 + O(NJ_N^3).$$

Both leading coefficients are strictly below $17/36$, which proves (5.1) for large N . The residue-restricted matching and the joint crooked/interval count are included in the formal proof. $\square$

Lean declaration: eventually_straddling_gap.

On the other hand, (4.4) gives

$$\frac{N^2}{2} \leq |\mathcal{G}| + 2048N. \quad (5.2)$$

For sufficiently large N , (5.1) contradicts (5.2). Consequently either c lies in every member, or an avoiding progression lies entirely to one side of c . Reflect the whole family if necessary so that one avoiding progression lies to the right.

For the nonempty avoiding family define

$$L = \max_{S \in \mathcal{G}, c \notin S} \min S, \quad R = \min_{S \in \mathcal{G}, c \notin S} \max S. \quad (5.3)$$

Pairwise intersection gives $L \leq R$, and the right-hand avoiding member gives $c < L$. Both extremes are attained: there are avoiding members P, T with $P \subseteq [L, N]$ and $T \subseteq [1, R]$. Every avoiding member has a point at most L and a point at least R . All members are nonempty by (5.2), once N is large.

Lean declaration: outside_member_envelope.

6. FINITE COPRIME DENSITIES AND THE MIDDLE REGION

The middle-region count must be summed over every step without leaving an uncontrolled multiple of N^2 . We therefore use an exactly normalized finite density rather than a limiting density with a fixed error. For $N \geq 1$ put

$$Z_N = \sum_{d=1}^N \frac{1}{d^2}, \quad \rho_N = Z_N^{-1}, \quad M_N = \sum_{d=1}^N \frac{\mu(d)}{d^2}. \quad (6.1)$$

Then

$$1 \leq Z_N \leq \frac{5}{3}, \quad \frac{3}{5} \leq \rho_N \leq 1, \quad \sum_{d=1}^N \frac{\rho_N}{d^2} = 1. \quad (6.2)$$

Lemma 6.1 (Finite normalization). *For $N \geq 1$,*

$$|M_N - \rho_N| \leq \frac{2H_N}{N}. \quad (6.3)$$

If the integers a_0, a_1, b_0, b_1 satisfy $0 \leq a_0 \leq a_1 \leq N$ and $0 \leq b_0 \leq b_1 \leq N$, then the number C of coprime pairs in $(a_0, a_1] \times (b_0, b_1]$ satisfies

$$|C - \rho_N(a_1 - a_0)(b_1 - b_0)| \leq 8(a_1 + b_1)H_N. \quad (6.4)$$

For integers $0 \leq a \leq b \leq N$, the number of coprime pairs $a < u < v \leq b$ is at most

$$\frac{\rho_N}{2}(b - a)^2 + 8bH_N. \quad (6.5)$$

Proof. Finite Möbius inversion gives

$$\sum_{dk \leq N} \frac{\mu(d)}{d^2 k^2} = 1.$$

The omitted terms in $M_N Z_N$ have total absolute value at most $2H_N/N$, by the tail estimate for $\sum k^{-2}$. Since $Z_N \geq 1$, division by Z_N gives (6.3).

For a rectangle $[1, A] \times [1, B]$, count common divisors:

$$\#\{(u, v) : 1 \leq u \leq A, 1 \leq v \leq B, (u, v) = 1\} = \sum_{d \leq \min(A, B)} \mu(d) \left\lfloor \frac{A}{d} \right\rfloor \left\lfloor \frac{B}{d} \right\rfloor.$$

Replacing floors and then replacing the truncated Möbius sum by ρ_N gives an error bounded by $2(A + B)H_N$. Inclusion-exclusion for the four corner rectangles yields (6.4). Splitting a square along its diagonal and using symmetry gives (6.5); the diagonal only decreases the required upper bound. $\square$

Lean declaration: `coprimeBox_density_error`.

6.1. The fixed-step count. Let the one-sided endpoints satisfy $c \leq L \leq R$. On the d -grid through c , put

$$C = \left\lfloor \frac{c}{d} \right\rfloor, \quad A = \left\lfloor \frac{L - c}{d} \right\rfloor, \quad M = \left\lfloor \frac{R - c}{d} \right\rfloor.$$

A rooted progression in the middle region has endpoint coordinates (l, r) with $1 \leq l \leq C$ and $A < r \leq M$. An intrinsic witness is assigned its gcd step d and primitive coordinates. Negative witnesses have one coordinate on either side of the root; positive witnesses have both coordinates to the right.

Let e be the largest right endpoint coordinate among the rooted progressions of this step, taking $e = A$ when there are none. A negative witness with right coordinate at most e cannot occupy an endpoint cell occupied by a progression. The two classes therefore fit together in the full rectangle of area $C(e - A)$. A positive witness cannot have its right coordinate at most e , because a rooted progression attaining e would contain its triple through c .

The remaining negative witnesses lie in the coprime rectangle $(0, C] \times (e, M]$. Positive witnesses with first coordinate at most e lie in $(0, e] \times (e, M]$; the rest lie in the coprime triangle $e < u < v \leq M$. Lemma 6.1 thus gives the joint bound

$$C(e - A) + \rho_N \left((M - e)(C + M) - \frac{(M - e)^2}{2} \right) + 40(C + M)H_N. \quad (6.6)$$

Privacy makes the witness encodings injective, while progression endpoints uniquely specify a progression. These observations justify counting the coordinate sets in place of the original members.

Lean declaration: `middle_coordinate_pools_card_1e`.

Converting (6.6) back to lengths gives, for a real $s_d \in [0, R - L]$, the following contribution from step d :

$$\frac{c(R - L - s_d) + \rho_N(s_d R - s_d^2/2)}{d^2} + (40H_N + 2) \frac{N}{d}. \quad (6.7)$$

The last term includes all rounding errors. The natural-grid conversion is uniform in c, L, R, d ; no divisibility of the endpoints is assumed.

Lean declaration: middle_nat_grid_scale.

7. ONE-SIDED COUNTING AND ENDPOINT STABILITY

Continue with the family and endpoints in (5.3). Restricted witnesses meeting $P \subseteq [L, N]$, followed by a choice of a shortest available witness in the exterior corner, give ordered witnesses $\{x, p\}$ for all but $N + E_N(c)$ crooked members, with

$$x < p, \quad x \leq R, \quad L < p, \quad x \neq L. \quad (7.1)$$

If $c < x < L$ and $p > R$, the crop $S \cap [x, p]$ is either $\{x, p\}$ or is crooked. Otherwise its arithmetic progression structure provides a shorter witness. The presence of $T \subseteq [1, R]$ rules out the need for a witness whose two entries are both greater than R .

For the progression members, discard the sets of size at most three and the exceptional fixed-endpoint classes. Their total cost is at most $2NH_N + 20N + 3$. The retained progressions have at least four points. A retained rooted progression has first endpoint less than c and last endpoint greater than L ; an avoiding progression has first endpoint at most L and last endpoint at least R .

7.1. The outer regions. Set $X = N - R$. Partition the prepared crooked members by the position of their ordered witness, and the prepared progressions by their step and right endpoint. In addition to the middle contribution (6.7), the following bounds apply:

Classes counted together	Upper bound
$x < c, p > R$, and rooted intervals ending after R	cX
$c < x < L, p > R$, and avoiding intervals ending after R	$(L - c)X + 2N$
$L < x \leq R < p$	$(R - L)X$
Progressions of step at least two, first endpoint at most L , last endpoint at least R	$\frac{2}{3}LX + 2NH_N + N$
Avoiding intervals with right endpoint exactly R	N

The first and third rows are counts in rectangular pair sets. The fourth is an endpoint count over steps $d \geq 2$, using $\sum_{d=2}^N d^{-2} \leq 2/3$. The fifth is an endpoint injection. For clarity, the joint count in the second row is described next.

Lemma 7.1 (Ordered rows). *The crooked members with $c < x < L < p$ and $p > R$, prepared as in (7.1), together with the avoiding intervals $[j, q]$ with $c < j \leq L \leq R < q \leq N$, number at most $(L - c)(N - R) + 2N$.*

Proof. In row i , let H_i count witnesses whose crop is exactly the pair, and let B_i count those with crooked crop. Let A_i count the avoiding intervals with left endpoint i , and put $J_i = B_i + A_i$. Privacy gives $H_i + B_i \leq X$. Call a row active when $A_i > 0$. For an active row let ℓ_i be the least right endpoint of an interval in that row. Crooked crops force all its interval endpoints to precede the first crooked-crop witness endpoint. Thus $J_i \leq N - \ell_i + 1 \leq X$.

If active rows satisfy $i < j$, each pair crop in row i must meet the interval $[j, \ell_j]$. Its right endpoint is therefore at most ℓ_j . Hence

$$H_i + J_j \leq X + 1.$$

Order the active rows increasingly. Pair the H term of each row with the J term of the next row; bound the first J and the last H separately. This gives at most $a(X+1)+X$ for a active rows. Inactive rows contribute at most X each. With $L - c$ rows in total, the bound is $(L - c)X + (L - c) + X \leq (L - c)X + 2N$. $\square$

Lean declaration: `corner_witnesses_and_intervals_card_le`.

7.2. Retaining the deficit. Adding the leading terms from the outer regions gives

$$\frac{5}{3}L(N - R) + (R - L)(N - R).$$

The exact normalization in (6.2) permits the following inequality after summing the middle contributions.

Lemma 7.2. *If $0 \leq c \leq L \leq R \leq N$ and $0 \leq s_d \leq R - L$, then*

$$\begin{aligned} & \frac{5}{3}L(N - R) + (R - L)(N - R) \\ & + \sum_{d=1}^N \frac{c(R - L - s_d) + \rho_N(s_d R - s_d^2/2)}{d^2} \leq \frac{N^2}{2} - \frac{L^2 + (N - R)^2}{100}. \end{aligned} \quad (7.2)$$

Proof. For $q, c, r, x \geq 0$, $0 \leq s \leq r$, $0 \leq k \leq 5/3$, and $n = q + c + r + x$, the elementary inequality used is

$$kx(q + c) + rx + kc(r - s) + s(q + c + r) - \frac{s^2}{2} \leq \frac{n^2}{2} - \frac{(q + c)^2 + x^2}{100}. \quad (7.3)$$

To verify it, put $L = q + c$ and $w = x + r - s$. The left side is at most $kLw + w^2/4 + s(L + w) + s^2/2$. Subtracting this from $n^2/2 = (L + w + s)^2/2$ leaves

$$\frac{L^2}{2} + \frac{w^2}{4} - (k - 1)Lw \geq \frac{L^2 + w^2}{100} \geq \frac{L^2 + x^2}{100}.$$

For the first inequality use $k \leq 5/3$ and the identity

$$\frac{49}{100}L^2 + \frac{6}{25}w^2 - \frac{2}{3}Lw = \frac{(7L - 5w)^2}{105} + \frac{7}{300}L^2 + \frac{1}{525}w^2.$$

Apply (7.3) with $q = L - c$, $r = R - L$, $x = N - R$, $s = s_d$ and $k = 5/3$. Multiply it by ρ_N/d^2 . Since $(5/3)\rho_N \geq 1$, its middle term bounds $c(R - L - s_d)/d^2$. Summing now uses $\sum_{d=1}^N \rho_N/d^2 = 1$ exactly and proves (7.2). $\square$

Lean declaration: `weighted_middle_deficit`.

Combining the table, (6.7), the preparation costs, and Lemma 7.2 gives the explicit estimate

$$\begin{aligned} |\mathcal{G}| & \leq \frac{N^2}{2} - \frac{L^2 + (N - R)^2}{100} \\ & \quad + (40H_N + 6)NH_N + 25N + 3 + E_N(c) \\ & \leq \frac{N^2}{2} - \frac{L^2 + (N - R)^2}{100} + 512NJ_N^3. \end{aligned} \quad (7.4)$$

The second inequality follows from $H_N \leq 2J_N$, $J_N \geq 1$, and (2.3). In particular, every error in the partition is included in a uniform term.

Lean declaration: `one_sided_family_natLog_bound`.

Proposition 7.3 (Endpoint stability). *Under (5.2) and the one-sided envelope conditions,*

$$(L + N - R)^2 \leq 512000NJ_N^3. \quad (7.5)$$

For every fixed $K > 0$, it follows that $K(L + N - R)J_N \leq N$ for all sufficiently large N .

Proof. Compare (7.4) with (5.2). Since $J_N \geq 1$,

$$L^2 + (N - R)^2 \leq 256000NJ_N^3.$$

Use $(L + N - R)^2 \leq 2(L^2 + (N - R)^2)$ to obtain (7.5). Then

$$(K(L + N - R)J_N)^2 \leq 512000K^2NJ_N^5 \leq N^2$$

eventually, because $J_N^5 = o(N)$. Taking nonnegative square roots proves the second assertion. In particular, $K = 256$ gives (1.3). $\square$

Lean declaration: `eventually_one_sided_endpoint_stability`.

8. THE AVOIDING PROGRESSIONS ARE INTERVALS

The endpoint estimate alone does not identify the avoiding progressions. The following extra gap completes that step.

Lemma 8.1. *Suppose c belongs to every crooked member of a well-intersecting family $\mathcal{G}$, every member has a point at most $L \leq N$, and $P \in \mathcal{G}$ avoids c and is a progression that is not an interval. For $N \geq 2$,*

$$|\mathcal{G}| \leq \frac{3}{8}N^2 + \frac{5}{3}LN + 2NH_N + 20N + 3 + E_N(c). \quad (8.1)$$

Proof. A noninterval progression has step at least two and cardinality at most $N/2 + 1$. Every crooked member meets P . Restricted witnesses therefore place all but $E_N(c)$ crooked members into pairs meeting P . The number of available pairs is at most $3N^2/8 + N$.

For a fixed step d , a progression with a point at most L has first endpoint at most L . Its endpoints lie in the same residue class as the other progressions of that step. Counting the endpoint rectangle and summing over d gives at most $(5/3)LN + 2NH_N + 19N + 3$ progression members, including the small exceptions. Addition gives (8.1). $\square$

Lean declaration: `noninterval_avoider_low_start_finite_bound`.

By Proposition 7.3, $256L \leq N$ eventually. Each rooted member has the point $c < L$, and each avoiding member has a point at most L by (5.3). If a noninterval avoiding progression existed, (8.1) would give

$$|\mathcal{G}| \leq \frac{7}{16}N^2$$

for all sufficiently large N . Here the coefficient $7/16$ leaves ample room for the term $(5/3)LN$ and for the error $O(NJ_N^3)$. This contradicts (5.2). Every avoiding member is therefore an ordinary interval $[a, b]$.

Its endpoints satisfy $a \leq L \leq R \leq b$. Since $c < L$ and the interval avoids c , necessarily $c < a$. We have proved the second alternative of Theorem 1.2, including (1.3). The first alternative was already separated before defining L, R . Together with (4.4), this proves that theorem.

Lean declaration: `structural_reduction`.

9. EXTERIOR PAIRS AND THE LINEAR CLOSING BOUND

Fix a well-intersecting family $\mathcal{G}$ satisfying the second alternative of Theorem 1.2. Put

$$\mathcal{Q} = \{\{u, v\} : c < u < L \leq R < v \leq N\}.$$

This is the exterior corner in which pairs will be reserved for the members avoiding c .

9.1. Witnesses outside reserved pairs. The following lemma is the geometric ingredient that turns the logarithmic stability estimate into a linear counting error.

Lemma 9.1 (Exterior witnesses). *Assume (1.3). Let $S \subseteq [N]$ be crooked and contain c . Let $\mathcal{W} \subseteq Q$ have the property that, for every $\{u, v\} \in \mathcal{W}$ with $u < v$, the set S has a point strictly between u and v . Then S has an intrinsic witness at c outside $\mathcal{W}$.*

Proof. Suppose all intrinsic witnesses lie in $\mathcal{W}$. In particular all lie in Q . A witness obtained from a left neighbor of c would contradict this, so $c = \min S$. The right neighbor of c must be $c + d < L$. Using that neighbor and the absence of witnesses outside the corner shows that the part of S up to R is a filled progression prefix

$$S \cap [1, R] = \{c, c + d, \dots, c + hd\}. \quad (9.1)$$

There is also a point of S beyond R ; otherwise S itself is a progression. Put $z = c + hd$.

The prefix reaches L . If it ended earlier, a tail point together with z would supply an intrinsic witness with no point of S between its entries, contradicting the assumption on $\mathcal{W}$. For each tail point $v > R$, the pair $\{z, v\}$ lies outside the corner and therefore has an arithmetic cover through c in S . Write e for its step. Its first point after c lies in the prefix, so $d \mid e$; therefore $d \mid (v - c)$. Its first point after z is $z + e$, which belongs to S and must exceed R . Since $e \leq z - c$, this gives $2z > R + c$. Thus all of S lies in the d -grid through c .

Let $v = \max S = c + md$. The coprime-index estimate in the prefix gives an integer j with

$$L \leq c + jd \leq z, \quad \gcd(j, m) = 1.$$

Here is the finite residue estimate. Set $t = L + N - R$, $a = \lfloor (L - c)/d \rfloor + 1$, and $q = 2a + \max(m - 2h, 0)$. If no integer in $[a, h]$ were coprime to m , reflection $j \mapsto m - j$ would put every reduced residue into the two end portions of length a or the middle interval $(h, m - h)$. Consequently $\varphi(m) \leq q$. The prefix bounds give $dq \leq 4t$, whereas $md > R - c \geq N - t$. The elementary inequality $m \leq (\lfloor \log_2 m \rfloor + 1)\varphi(m)$ would then imply

$$N - t < md \leq dqJ_N \leq 4tJ_N \leq N/64,$$

contradicting $t \leq N/256$. This proves the required existence of j .

Since $c + jd \geq L$, the pair $\{c + jd, v\}$ also has an arithmetic cover in S . Its step e divides both jd and md , hence divides d . Conversely the first step of that cover lies in the filled prefix, so $d \mid e$. Thus $e = d$, and its cover fills the entire d -grid from c to v . This is all of S , a contradiction. $\square$

Lean declaration: exists_witness_outside_exterior_pairs.

9.2. Matching around the corner. The progression matching can also avoid the whole exterior corner while retaining its $20000N$ exception bound. Here is the normalized construction. For step d write

$$A = \left\lfloor \frac{L - c}{d} \right\rfloor + 1, \quad B = \left\lfloor \frac{R - c}{d} \right\rfloor, \quad C = \left\lfloor \frac{c}{d} \right\rfloor.$$

For each right endpoint row $r > B$, the long-core condition gives

$$A + C + 1 \leq \varphi(r) \quad \text{when } d \leq L + N - R. \quad (9.2)$$

Indeed, $r \leq J_N \varphi(r)$, whereas $rd > R - c$ and $256(L + N - R)J_N \leq N$. The floor errors in $A + C + 1$ are absorbed by this last inequality.

At most A reduced residues are lost by requiring $a \geq A$. Hence (9.2) assigns all $C + 1$ possible left endpoints injectively to positive primitive pairs (a, r) with $a \geq A$. These pairs lie to the right of the forbidden corner's first-coordinate range. For shorter rows use the matching of Lemma 3.1; their second coordinates are at most B , so they cannot lie in the corner or collide with the new long-row pairs. If $d > L + N - R$, the forbidden first-coordinate interval contains no nonzero grid point, and the original matching already avoids the corner. This constructs the required matching for all d .

Lean declaration: exists_natural_corner_pair_matching.

Combine this matching with Lemma 9.1 and with the three-element assignment from Proposition 3.2. The latter cannot occupy a reserved pair: a set consisting of c and that pair has no point strictly between its entries. We obtain

$$|\{S \in \mathcal{G} : c \in S\}| \leq \binom{N}{2} - |\mathcal{W}| + 20001N, \quad (9.3)$$

provided every rooted member has a point between the entries of each reserved pair in $\mathcal{W}$.

Lean declaration: `star_card_le_with_reserved_pairs`.

Proposition 9.2 (Long-core bound). *Every family in the second alternative of Theorem 1.2 satisfies $|\mathcal{G}| \leq \binom{N}{2} + 20003N$.*

Proof. Set aside the avoiding intervals with left endpoint $c + 1$ or right endpoint N ; there are at most $2N$ of them. For every other avoiding interval $[a, b]$, reserve the pair $\{a - 1, b + 1\}$. These pairs lie in $\mathcal{Q}$ and are distinct, since they recover the interval. Every rooted member meets $[a, b]$, so it has a point strictly between $a - 1$ and $b + 1$. Apply (9.3) to the set $\mathcal{W}$ of all reserved pairs. The $-|\mathcal{W}|$ term pays for all the nonboundary avoiding intervals. Adding the at most $2N$ boundary intervals gives the result. $\square$

Lean declaration: `long_core_family_card_le`.

Proof of Theorem 1.1. Choose N large enough for Theorem 1.2. If $|\mathcal{F}| < N^2/2$, the desired upper bound is immediate. Otherwise obtain $\mathcal{G}$ from that theorem. Proposition 3.2 in the first case, and Proposition 9.2 in the second, give

$$|\mathcal{F}| \leq \binom{N}{2} + (20003 + 2048)N = \binom{N}{2} + 22051N \leq \frac{N^2}{2} + 30000N.$$

The lower construction in Section 1 gives $t(N) - N^2/2 \geq 1 - N/2$, so the two-sided big- O conclusion follows. $\square$

10. THE LEAN FORMALIZATION

The accompanying file `Solution.lean` contains the complete formal proof. It imports only `FormalConjectures.ErdosProblems.«272»` and `TaskSupport`. The final theorem is `Bounty.target`. Its type, obtained from the source declaration by the `fcTypeOfName%` macro, is the assertion

$$(N \mapsto t(N) - N^2/2) = O[\text{atTop}](N \mapsto N),$$

with $t(N)$ represented by `Erdos272.maxArithInterCard` and the casts to $\mathbb{R}$ made explicit in Lean. The macro reads the source theorem's type; the proof does not use that theorem as a dependency.

The final assembly in the namespace `Bounty` is:

```
theorem target :
  fcTypeOfName% "Erdos272.erdos_272.variants.szabo_strong" := by
  exact target_of_structural_reduction structural_reduction
```

Here `structural_reduction` is a proved theorem. Its hypotheses do not include an unproved structural conjecture. The conditional assembly lemma preceding it is discharged by the completed reduction.

The proof uses finite sets of finite sets of natural numbers. Arithmetic progressions are related explicitly to `Mathlib's Set.IsAPOfLength`, and integer-centered models are transported back to the natural-number ground set. Primitive-pair assignments are partial injections with a bounded number of dummy values. All cardinality inequalities are finite; the final use of asymptotics converts uniform eventual bounds to the task's big- O statement.

10.1. Verification record. The standalone file has 12,797 lines and 519 named theorems. It compiles under Lean 4.33.1, with Formal Conjectures commit

8432eac998110a563e03df65a28c117e97c8c142

and Mathlib commit

0df444a360eaa60ab8c11dca51a86af692955474.

Every named theorem passed an axiom audit. The final theorem uses only `propext`, `Classical.choice`, and `Quot.sound`, the three axioms permitted by the task. A traversal of 45,421 transitive declaration dependencies found neither an admission nor the forbidden source theorem. The final type expression is identical to the source type; its canonical SHA-256 matches the task manifest:

372a052300899d71f5b27d23e5505351f9a1400793e425b51628356a0739daf6

The formal certificate is independent of the prose presentation: these checks certify the Lean statement and proof. The source-policy check from the official task validator also passed. The full production validator, including its Linux sandbox and comparator/export executables, was not run in the Windows development environment.

10.2. Files and reproducibility. The proof package contains:

- `Solution.lean`, the complete standalone proof;
- `submission/Main.lean`, the identical proof body for the task’s upload interface, which supplies its fixed imports and namespace wrapper automatically;
- `verification.json`, the compilation, type, dependency, axiom, and static-policy verification record;
- the LaTeX source and PDF of this paper.

Concatenating the trusted task header, `submission/Main.lean`, and the trusted footer reproduces `Solution.lean` byte for byte. Its SHA-256 is

0d651c16942333977ee1ee4ddf17d017e3510f0dc6f2f1dd3542e434d16eb91e

In the pinned task environment the standalone proof is checked by `lake env lean Solution.lean`. This command presupposes the task’s `TaskSupport` module and pinned dependencies; a different Mathlib revision is not a substitute for that environment.

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The authors used OpenAI’s Codex extensively in developing and integrating the mathematical arguments, constructing and debugging the Lean formalization, checking source material, and preparing this manuscript. The AI contribution included proof development and formal verification work, as well as exposition and typesetting. The human authors are responsible for the mathematical claims, the final text, and any submission of the paper. Mechanical verification refers specifically to the Lean certificate and the checks described in Section 10.

APPENDIX A. A GUIDE TO THE FORMAL CERTIFICATE

All declarations listed below are in the namespace `Bounty`. The names locate the finite arguments supporting the corresponding parts of the paper. They are included to make the longer combinatorial bookkeeping directly inspectable.

Mathematical component	Principal declarations
Lower construction	<code>lowerFamily_admissible</code> ; <code>card_lowerFamily</code> ; <code>lower_bound</code>
Rooted pair matching	<code>exists_progression_pair_matching</code> ; <code>exists_natural_progression_pair_matching</code> ; <code>common_point_card_le</code>
Dense crooked members	<code>dense_crooked_card_le</code>
Restricted witnesses	<code>witnessExceptionBound_le_natLog</code>

Mathematical component	Principal declarations
Size bands and sparse root	<code>crooked_size_band_card_mul_le;</code> <code>crooked_no_common_point_card_mul_le;</code> <code>eventually_sparse_crooked_common_point</code> <code>eventually_common_crooked_reduction</code>
Linear discard	
Straddling exclusion	<code>two_step_straddling_family_card_le;</code> <code>large_step_straddling_family_card_le;</code> <code>eventually_straddling_gap</code> <code>outside_member_envelope</code>
Avoiding envelope	
Finite coprime normalization	<code>finiteCoprimeDensity_weights_sum;</code> <code>coprimeBox_density_error;</code> <code>coprimeUpperTriangle_card_le</code>
Middle count	<code>middle_coordinate_pools_card_le;</code> <code>middle_nat_grid_scale;</code> <code>middle_prepared_family_bound</code>
Joint row count	<code>corner_witnesses_and_intervals_card_le</code>
One-sided deficit	<code>retained_deficit;</code> <code>weighted_middle_deficit;</code> <code>one_sided_family_natLog_bound</code>
Stability and intervals	<code>eventually_one_sided_endpoint_stability;</code> <code>eventually_low_start_interval_avoiders</code>
Complete reduction	<code>eventually_structural_reduction;</code> <code>structural_reduction</code>
Reserved exterior pairs	<code>exists_witness_outside_exterior_pairs;</code> <code>exists_natural_corner_pair_matching;</code> <code>star_card_le_with_reserved_pairs</code>
Final finite bound	<code>long_core_family_card_le;</code> <code>finite_upper_bound_of_structural_reduction</code>
Exact task theorem	<code>target</code>

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